

SUMMER INTERNSHIP

PRESENTATION REPORT

C++ Programming Internship

Console Applications, OOP & File Handling in C++

STUDENT CREDENTIALS

Student Name Ravi Shankar kumar

Enrollment / Roll No. 25SCS1003005205

Program & Semester 3rd Semester

Host Organization Thiranex – Skill Development & Future Tech

Internship Duration 30 Jul 2026 – 29 Aug 2026

Intern ID THX-JUL3026-505

Candidate & Program Overview

Key details of the student and internship program

CANDIDATE DETAILS

Student Name: Ravi Shankar kumar
Roll / Enrollment No.: 25SCS1003005205
Semester: 3rd Semester
University: IILM University, Greater Noida, U.P.

INTERNSHIP PROFILE

Role Offered: Intern – C++ Programming
Host Organization: Thiranex – Skill Development & Future Tec
Duration: 30 Jul – 29 Aug 2026 (~4.5 Weeks)
Intern ID: THX-JUL3026-505

Internship at a Glance

A quick snapshot of the internship engagement

01 ORGANIZATION

Thiranex

02 DURATION

30 Jul – 29 Aug 2026

03 DOMAIN

C++ Programming

04 MODE

Remote / Project-Based

ABOUT THE ROLE

Undertook a structured, milestone-based C++ Programming internship covering problem-solving, object-oriented programming and file handling thr

Internship Offer Letter

Formal offer details

OFFER DETAILS

Intern Name: Ravi Shankar kumar

Intern ID: THX-JUL3026-505

Internship Role: Intern – C++ Programming

Commencement Date: 30 Jul 2026

Completion Date: 29 Aug 2026

Work Mode: Remote / Project-Based

About the Organization

Thiranex — Skill Development & Future Tech

ORGANIZATION OVERVIEW

Thiranex provides structured, project-based skill development and future-technology in

KEY POINTS

- ✓ Structured, milestone-based internship tracks defined
- ✓ Project-driven learning
- ✓ Defined deliverables and deadlines
- ✓ Online progress tracking
- ✓ Certification on successful completion

Objectives of the Internship

What the internship set out to achieve

01 Master Core C++ Concepts

Apply structures, classes and OOP principles.

02 Implement File Handling

Use file I/O to persist records across runs.

03 Build Menu-Driven Applications

Design clean console interfaces for applications.

04 Apply Programming Logic

Use loops, arrays and conditions for interactive programs.

Module 1: Student Management System

Completed Successfully

TASK DESCRIPTION

Developed a console-based Student Management System in C++ using file handling and structured programming.

EXPECTED OUTCOME

A functional console application demonstrating practical C++ programming, logic, structured design and reliable data handling.

KEY FEATURES IMPLEMENTED

- ✓ Add, update, delete, and display student information
- ✓ Persistent file storage
- ✓ Menu-driven navigation
- ✓ Input validation

✓ COMPLETED SUCCESSFULLY

Module 2: Bank Management Application

Completed Successfully

TASK DESCRIPTION

Designed a C++ banking application simulating core operations using OOP and file ha

KEY FEATURES IMPLEMENTED

- ✓ Deposit, withdrawal and balance check
- ✓ Classes and objects
- ✓ Persistent customer data
- ✓ Transaction input validation

EXPECTED OUTCOME

A functional console application demonstrating practical C++ programming, logic, structured design and reliable data handling.

✓ COMPLETED SUCCESSFULLY

Module 3: Library Management System

Completed Successfully

TASK DESCRIPTION

Developed a Library Management System for books, members and borrowing records

EXPECTED OUTCOME

A functional console application demonstrating practical C++ programming, logic, structured design and reliable data handling.

KEY FEATURES IMPLEMENTED

- ✓ Book issue and return
- ✓ Search by title or author
- ✓ Structures/classes for records
- ✓ File-based persistence

✓ COMPLETED SUCCESSFULLY

Module 4: Mini Game Project

Completed Successfully

TASK DESCRIPTION

Created a console-based mini game demonstrating loops, arrays and conditional logic.

EXPECTED OUTCOME

A functional console application demonstrating practical C++ programming, logic, structured design and reliable data handling.

KEY FEATURES IMPLEMENTED

- ✓ Dynamic game board
- ✓ Win/loss detection
- ✓ Replay option
- ✓ Clean interactive game logic

✓ COMPLETED SUCCESSFULLY

Tools & Technologies Used

The stack applied across the internship

01 C++ Language

Core syntax, control structures, functions

02 Object-Oriented Programming

Classes, objects, encapsulation

03 File Handling

fstream-based persistent data storage

04 Structures & Arrays

Structured records and data collections

05 STL Concepts

Vectors, strings and standard containers

06 Compiler / IDE

GCC / Code::Blocks / Dev-C++

Learning Outcomes & Skills Gained

Technical and professional takeaways

CORE C++ PROGRAMMING

- ✓ Structures, classes & OOP design
- ✓ Control flow, loops & arrays
- ✓ Functions & modular code organization

DATA PERSISTENCE

- ✓ File handling with fstream
- ✓ Persistent records
- ✓ Data validation & error handling

APPLICATION DESIGN

- ✓ Menu-driven console interfaces
- ✓ CRUD-style record management
- ✓ Interactive program/game logic

Challenges & Key Learnings

Reflections from the internship journey

! Managing file I/O reliably

Learned to handle file-open failures and preserve records correctly.

! Structuring OOP designs

Improved from procedural code toward encapsulated classes and objects.

! Designing clean menu systems

Refined validation and loop-driven menus for smoother interaction.

! Implementing game logic

Learned to manage game state, conditions and win/loss logic cleanly.

Certificate of Completion

Official recognition of successful internship completion

CERTIFICATE DETAILS

Certificate ID: THX-JUL3026-505

Issued for successful completion of C++ Programming internship

Duration: 30 Jul 2026 – 29 Aug 2026

Signed by: Hariharan M, Founder & CEO

Conclusion

Progression through core C++ programming, including console applications, object-oriented programming, file handling, and database operations.

4/4

Milestones Completed

100%

On-Track Submissions

4

C++ Applications Built

THANK YOU

Ravi Shankar kumar | 3rd Semester | IILM University, Greater Noida